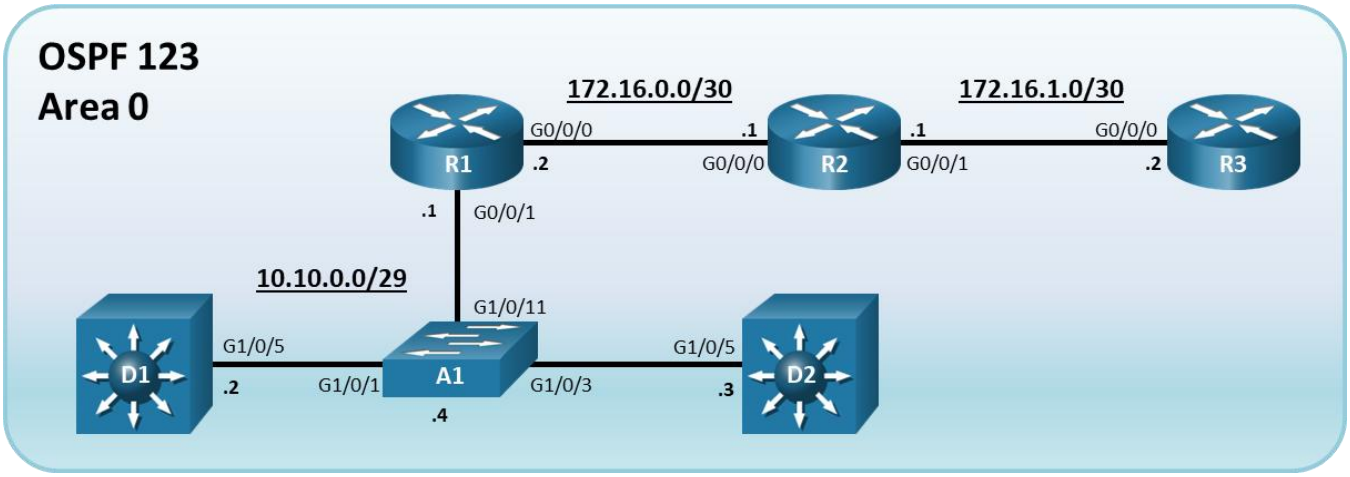
Lab - Implement NTP

Topology



Addressing Table

Device	Interface	IPv4 Address
R1	G0/0/0	172.16.0.2/30
	G0/0/1	10.10.0.1/29
R2	G0/0/0	172.16.0.1/30
	G0/0/1	172.16.1.1/30
R3	G0/0/0	172.16.1.2/30
D1	G1/0/5	10.10.0.2/29
D2	G1/0/5	10.10.0.3/29
A1	VLAN 1	10.10.0.4/29

Objectives

Part 1: Build the Network, Configure Basic Device Settings and Routing

Part 2: Configure NTP in a P2P Network

Part 3: Configure NTP in a Multiaccess Broadcast Network

Background / Scenario

Accurate and synchronized time is crucial for the efficient operation of network devices, enabling effective troubleshooting, precise Syslog and SNMP event reporting, and the correlation of security events across multiple systems. While devices use either software clocks (which reset on reboot) or hardware clocks (which retain time), neither inherently provides synchronization across the network. Network Time Protocol (NTP)

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offers a scalable solution by synchronizing device clocks through a hierarchical architecture, where clients periodically poll authoritative NTP servers, using multiple messages to ensure high accuracy.

Note: NTP uses UDP port 123 and is documented in RFC 1305. Current versions include NTPv3 and NTPv4.

NTP is commonly implemented in three ways depending on the network type:

- **Point-to-point (P2P)** - Consists of an NTP server (master) and NTP client.
- **Broadcast** - Consists of an NTP server broadcasting (i.e., multicasting) to listening NTP broadcast clients. The server is configured using the **ntp broadcast** interface command and clients are configured using the **ntp broadcast client** interface command.
- **NTP Peers** - Used in larger redundant topologies for backup purposes where two NTP servers are also peers to query different external time sources. The peers establish a client server relationship with each other and try to offer their clock settings to each other. NTP peers are configured using the **ntp peer** *peer-ip-address* global config command.

In this lab, you will configure NTP as follows:

- R3 will be the designated authoritative time source.
- R2 and R3 will establish a P2P client-server NTP connection while R1 and R2 will establish a P2P client-server NTP connection.
- R1 will be configured as an NTP broadcast server to the NTP broadcast clients (i.e., A1, D1, and D2).

Note: This lab is an exercise in developing, deploying, and verifying how NTP operates and does not reflect networking best practices.

Note: The routers used with this CCNP hands-on lab are Cisco 8200L and the two switches are Cisco 9300L and 9200L switches. Other routers and switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no start-up configurations. If you are unsure contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS release 17.12.05 universal image or comparable)
- 1 PC (Choice of operating system with a terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network, Configure Basic Device Settings and Routing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on the routers, Layer 3 switches, and the Layer 2 switch. You will also configure multiarea OSPFv2 on the routers and Layer 3 switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

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Step 2: Configure basic settings for the routers.

a. Console into each router, enter global configuration mode, and apply the basic settings, interface addressing, and OSPFv2 configuration. A command list for each router is provided below for initial configurations.

Note: Routers were configured with OSPFv2 using the interface configuration method.

Router R1

```
hostname R1
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is R1, Implement NTP Lab #
interface g0/0/0
 ip add 172.16.0.2 255.255.255.252
 ip ospf 123 area 0
 no shut
 exit
interface GigabitEthernet0/0/1
 ip address 10.10.0.1 255.255.255.248
 ip ospf 123 area 0
 no shut
 exit
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is R2, Implement NTP Lab #
interface g0/0/0
 ip add 172.16.0.1 255.255.255.252
 ip ospf 123 area 0
 no shut
 exit
interface GigabitEthernet0/0/1
 ip address 172.16.1.1 255.255.255.252
 ip ospf 123 area 0
 no shut
 exit
```

Router R3

```
hostname R3
```

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```
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is R3, Implement NTP Lab #
interface g0/0/0
 ip address 172.16.1.2 255.255.255.252
 ip ospf 123 area 0
 no shut
 exit
```

b. Save the running configuration to startup-config.

Step 3: Configure basic settings for the switches.

a. Console into each switch, enter global configuration mode, and apply the basic settings and interface addressing. A command list for each switch is provided below for the initial configurations.

Note: Switches were configured with OSPFv2 using the interface method.

Switch D1

```
hostname D1
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
banner motd # This is D1, Implement NTP Lab #
ip routing
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface g1/0/5
 no switchport
 ip address 10.10.0.2 255.255.255.248
 ip ospf 123 area 0
 no shut
 exit
```

Switch D2

```
hostname D2
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is D2, Implement NTP Lab #
ip routing
interface range g1/0/1-24, g1/1/1-4, g0/0
```

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```
 shutdown
interface g1/0/5
 no switchport
 ip address 10.10.0.3 255.255.255.248
 ip ospf 123 area 0
 no shut
 exit
```

Switch A1

```
hostname A1
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
banner motd # This is A1, Implement NTP Lab #
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface range g1/0/1, g1/0/3, g1/0/11
 no shutdown
interface vlan 1
 ip address 10.10.0.4 255.255.255.248
 no shut
 exit
ip default-gateway 10.10.0.1
```

b. Save the running configuration to startup-config.

Step 4: On D1, verify end-to-end connectivity and time.

a. From D1, **ping** the R3 IP address (i.e., 172.16.1.2).

```
D1# ping 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/8 ms
```

There is end-to-end connectivity up to R3.

b. Verify the time on D1 using the **show clock** command.

```
D1# show clock
*00:27:00.528 UTC Mon Mar 1 1993
```

Notice the time reference on D1 is not accurate. Newer routers, including 8200 routers and 9300 switches, have a hardware clock. Your router may have the correct time. Optionally, you could use the **clock** command to change the software clock to the incorrect time to view the impact of using NTP.

```
D1# clock set 00:00:00 1 Mar 1993
*Mar  1 00:00:00.040: %SYS-6-CLOCKUPDATE: System clock has been updated from 19:54:00
UTC Wed Oct 29 2025 to 00:00:00 UTC Mon Mar 1 1993, configured from console by
console.
```

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```
Mar  1 00:00:00.040: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been set.
D1# show clock
00:00:04.452 UTC Mon Mar 1 1993
```

Step 5: On D2, verify end-to-end connectivity and time.

a. From D2, **ping** the R3 IP address (i.e., 172.16.1.2).

```
D2# ping 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/8 ms
```

There is end-to-end connectivity up to R3.

b. Verify the time on D2 using the **show clock** command.

```
D2# show clock
*00:29:25.759 UTC Mon Mar 1 1993
```

The time on D2 is also not accurate. If it is correct, you can change it with the **clock** command.

Step 6: On A1, R1 and R2, verify end-to-end connectivity and time.

a. From A1, R1, and R2 verify that they have connectivity to R3.

b. Verify their time.

Part 2: Configure NTP in a P2P Network

In this part, NTP will be configured for P2P. Specifically, R3 will be the authoritative server for R2, and R2 will be the server for R1.

The device selected as the authoritative time server is configured using the **ntp master** *stratum* global configuration command. The *stratum* variable identifies how many hops the server is away from an atomic time source.

The NTP client identifies the NTP server to associate with using the **ntp server** *server-ip-address* [**prefer**] [**source** *interface-id*] global configuration command. For redundancy purposes, multiple NTP servers can be specified using separate **ntp server** command. Use the **prefer** option to peer with a specific NTP server. Optionally, the source interface can be configured to stipulate the source IP address for queries for that server.

Step 1: On R3, configure the system clock.

In this lab, R3 is configured as the authoritative time source.

a. Verify the system clock on R3.

```
R3# show clock
*21:45:50.081 UTC Mon Jan 20 2020
```

If they clock is correct on your device, because it has a hardware clock, assume it is not for the purposes of this step.

b. On R3, manually reconfigure the system clock using the **clock set** privileged EXEC mode command. The time you set should be the Coordinated Universal Time (UTC) value. For example, the clock is set assuming that the current UTC is 7:25 pm (i.e., 19:25).

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```
R3# clock set ?
  hh:mm:ss  Current Time

R3# clock set 15:00:00 ?
  <1-31>  Day of the month
  MONTH   Month of the year

R3# clock set 15:00:00 20 ?
  MONTH  Month of the year

R3# clock set 15:00:00 29 Oct 2025
R3#
*Oct 29 15:00:00.054: %SYS-6-CLOCKUPDATE: System clock has been updated from 20:07:07
UTC Wed Oct 29 2025 to 15:00:00 UTC Wed Oct 29 2025, configured from console by
console.
Oct 29 15:00:00.064: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been set.
```

An informational message is generated stating that the system clock has been altered on the console.

c. Verify that the system clock has been updated.

```
R3# show clock
15:02:51.212 UTC Wed Oct 29 2025
```

Note: The default time zone is UTC.

d. Next, we will change the default time zone from UTC to Eastern Standard Time (EST) by specifying a -5 hours-offset between UTC and EST using the **clock timezone** *WORD hours-offset* global configuration command.

```
R3# config t
R3(config)# clock timezone ?
  WORD  name of time zone
```

The *WORD* can be anything you want. In our example, we'll use EST for Eastern Standard Time in the United States.

Note: Time zones have a set number of offset hours that must be specified to adjust from UTC.

```
R3(config)# clock timezone EST -5
R3(config)#
Oct 29 15:03:38.585: %SYS-6-CLOCKUPDATE: System clock has been updated from 15:03:38
UTC Wed Oct 29 2025 to 10:03:38 EST Wed Oct 29 2025, configured from console by
console.
R3(config)# exit
```

e. Verify how the system clock has been updated using the **detail** keyword.

```
R3# show clock detail
10:04:58.257 EST Wed Oct 29 2025
Time source is user configuration
```

Notice how the time and time zone have changed to reflect the configured command. It also states how the time was adjusted.

Note: Other time related commands that can be configured include the **clock summer-time** command to automatically switch between standard time and daylight savings time.

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Step 2: Configure R3 to be the authoritative time source.

Setting the clocks manually is not an accurate method of tracking time and events in networks. Manually configuring the time on all network devices is also not a scalable solution.

NTP enables network devices to poll an authoritative time source to ensure that all network devices are synchronized.

Time accuracy is commonly sourced from an external source, such as an atomic clock or a GPS receiver. Another option is to configure a device to be an authoritative time source. However, this command should only be used if you do not have a reliable external reference clock.

In this lab, R3 will be configured as an authoritative time source using the **ntp master** *stratum* command. The stratum number should be configured with a high number if a more reliable NTP source becomes available. When multiple NTP servers are specified, an NTP-enabled client automatically chooses the server with the lowest stratum number as its time source.

Note: The lower the stratum number the more trustworthy the accuracy of the time source.

a. Configure R3 as the authoritative time source with a stratum of 10 using the **ntp master** command.

```
R3(config)# ntp master ?
  <1-15>  Stratum number
  <cr>    <cr>

R3(config)# ntp master 10
R3(config)# exit
```

b. Verify if there are any NTP clients or peers on the network using the **show ntp associations** command.

```
R3# show ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~127.127.1.1     .LOCL.           9     14     16     1  0.000   0.000 7937.9
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

There are no other NTP clients. The address **127.127.1.1** is the loopback IP address of R3 that was assigned by the **ntp master** command. The reference clock is LOCL (i.e., local) with a stratum number of 9 which is one less than the configured number of 10.

c. Verify the status of NTP using the **show ntp status** command.

```
R3# show ntp status
Clock is synchronized, stratum 10, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10
ntp uptime is 2500 (1/100 of seconds), resolution is 4000
reference time is E1D080F7.F6041B38 (14:41:43.961 EST Mon Jan 20 2020)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 3939.38 msec, peer dispersion is 3938.29 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 16, last update was 9 sec ago.
```

Because R3 is the NTP master, it is synchronized with itself (i.e., 127.127.1.1).

Note: It may take several minutes before the clock is synchronized with itself.

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Step 3: Configure R2 to be an NTP client.

In this step, R2 will acquire its time source from R3. Therefore, R2 is an NTP client and will identify the NTP server to associate with using the **ntp server** *server-ip-address* [**prefer**] [**source** *interface-id*] global configuration command.

a. Verify that the system clock on R2.

```
R2# show clock
*22:16:10.632 UTC Sun Jan 12 2020
```

The time is inaccurate.

b. Configure R2 to synchronize to R3 using the **ntp server** *server-ip-address* command.

```
R2(config)# ntp server 172.16.1.2
R2(config)#
Oct 29 15:08:40.684: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been set.
```

c. The local time zone must again be adjusted.

```
R2(config)# clock timezone EST -5
R2(config)#
Oct 29 15:09:28.352: %SYS-6-CLOCKUPDATE: System clock has been updated from 15:09:28
UTC Wed Oct 29 2025 to 10:09:28 EST Wed Oct 29 2025, configured from console by
console.
R2(config)# exit
```

d. Now verify the local time on R2.

```
R2# show clock detail
10:10:01.911 EST Wed Oct 29 2025
Time source is NTP
```

Notice that the time and time zone are accurate. The output also confirms that NTP was the time source.

e. Verify if there are any NTP clients or peers on the network using the **show ntp associations** command.

```
R2# show ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~172.16.1.2      127.127.1.1     10     91    128    37  1.000  -2.500  2.631
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

The output confirms that R2 has associated with R3 (i.e., 172.16.1.2). Notice how it also identified the source of the R3 NTP information as 127.127.1.1 at stratum 10.

f. Verify the status of NTP using the **show ntp status** command.

```
R2# show ntp status
Clock is synchronized, stratum 11, reference is 172.16.1.2
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10
ntp uptime is 47200 (1/100 of seconds), resolution is 4000
reference time is E1D08495.D70A3FC0 (14:57:09.840 EST Mon Jan 20 2020)
clock offset is -2.5000 msec, root delay is 1.00 msec
root dispersion is 11.36 msec, peer dispersion is 2.63 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000000006 s/s
system poll interval is 128, last update was 114 sec ago.
```

The output confirms that R2 is synchronized with R3. When synchronized, R2 also becomes a stratum 11 NTP server.

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Note: It may take a few minutes before the time is synchronized with the NTP server.

Step 4: Configure R1 to be an NTP client.

In this step, R1 will become an NTP client of R2. No other configuration is required on R2 for it to advertise itself as an NTP server. However, R1 must identify R2 as its service using the **ntp server** command.

a. Verify the system clock on R1.

```
R1# show clock
*13:54:28.867 UTC Mon Jan 20 2020
```

The time is inaccurate.

b. Configure R1 to synchronize with R2 using the **ntp server** command.

```
R1(config)# ntp server 172.16.0.1
R1(config)#
Oct 29 15:11:34.487: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been set.
```

c. The local time zone must again be configured.

```
R1(config)# clock timezone EST -5
R1(config)#
Oct 29 15:11:43.925: %SYS-6-CLOCKUPDATE: System clock has been updated from 15:11:43
UTC Wed Oct 29 2025 to 10:11:43 EST Wed Oct 29 2025, configured from console by
console.
R1(config)# end
```

d. Verify the local time on R1.

```
R1# show clock detail
10:12:28.312 EST Wed Oct 29 2025
Time source is NTP
```

Notice that the time and time zone are accurate. The output also confirms that NTP was the time source.

Note: It may take a few minutes before the time is updated with the correct time.

e. Verify if there are any NTP clients or peers on the network using the **show ntp associations** command.

```
R1# show ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~172.16.0.1      172.16.1.2      11     10     64     1  1.000   0.500 189.44
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

The output confirms that R1 is a client with R2. It also identifies R3 (i.e., 172.16.1.2) which is 11 hops away as the source of R2 NTP information.

f. Verify the status of NTP using the **show ntp status** command.

```
R1# show ntp status
Clock is synchronized, stratum 12, reference is 172.16.0.1
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10
ntp uptime is 3200 (1/100 of seconds), resolution is 4000
reference time is E1D08C96.116872E0 (15:31:18.068 EST Mon Jan 20 2020)
clock offset is 0.5000 msec, root delay is 1.00 msec
root dispersion is 210.16 msec, peer dispersion is 189.44 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 64, last update was 19 sec ago.
```

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The output confirms that R1 is synchronized with R2. It now becomes a stratum 12 NTP server.

Note: It may take a few minutes before the time is synchronized with the NTP server.

Part 3: Configure NTP in a Multiaccess Broadcast Network

In this part, NTP will be configured in a multiaccess broadcast network. Specifically, R1 will be the NTP server for the A1, D1, and D2 Ethernet clients.

In a multiaccess network, an NTP server can broadcast (i.e., multicast) NTP updates to listening NTP broadcast clients.

The NTP server Ethernet interface is configured using the **ntp broadcast** interface command.

NTP clients are configured using the **ntp broadcast client** interface command.

Step 1: Configure R1 to be an NTP broadcast server.

In this step, R1 will broadcast NTP advertisements to NTP clients using the **ntp broadcast** command on its GigabitEthernet 0/0/1 interface.

Enter the G0/0/1 interface and configure R1 to be an NTP broadcast server using the **ntp broadcast** command.

```
R1(config)# interface g0/0/1
R1(config-if)# ntp broadcast
R1(config-if)# end
```

Step 2: Configure A1 as an NTP broadcast client.

In this step, A1 will be enabled as an NTP broadcast client using the **ntp broadcast client** command on its management interface.

a. Enter the VLAN management interface for of A1 (i.e., VLAN 1) and enable it as an NTP broadcast client.

```
A1(config)# interface vlan 1
A1(config-if)# ntp broadcast client
A1(config-if)# exit
```

b. Configure the local time zone.

```
A1(config)# clock timezone EST -5
A1(config)#
*Oct 29 20:15:31.674: %SYS-6-CLOCKUPDATE: System clock has been updated from 20:15:31
UTC Wed Oct 29 2025 to 15:15:31 EST Wed Oct 29 2025, configured from console by
console.
A1(config)# end
```

Notice the informational message.

c. Verify the local time on A1.

```
A1# show clock detail
*10:15:08.239 EST Wed Oct 29 2025
Time source is NTP
```

Notice that the time and time zone are accurate. The output also confirms that NTP was the source of the time.

d. Verify if there are any NTP clients or peers on the network using the **show ntp associations** command.

```
A1# show ntp associations
```

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```
  address          ref clock       st   when   poll reach   delay   offset    disp
* 10.10.0.1        172.16.0.1      12     43     64   377  -8.633   44.242   1.917
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

The output confirms that A1 is a client with R1 (i.e., 10.10.0.1).

e. Verify the status of NTP using the **show ntp status** command.

```
A1# show ntp status
Clock is synchronized, stratum 13, reference is 10.10.0.1
nominal freq is 119.2092 Hz, actual freq is 119.2090 Hz, precision is 2**18
ntp uptime is 121300 (1/100 of seconds), resolution is 8403
reference time is E1D093BE.047BFDC7 (16:01:50.017 EST Mon Jan 20 2020)
clock offset is 45.1660 msec, root delay is -6.65 msec
root dispersion is 91.91 msec, peer dispersion is 1.92 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000001692 s/s
system poll interval is 64, last update was 50 sec ago.
```

The output confirms that A1 is synchronized with R1 and is now a stratum 13 NTP server.

Note: It may take a few minutes before the time is synchronized with the NTP server.

Step 3: Configure D1 and D2 as NTP broadcast clients.

In this step, D1 and then D2 will be enabled as NTP broadcast clients using the **ntp broadcast client** command on their G1/0/5 interfaces.

a. On D1, enter interface g1/0/5 and enable it to be an NTP broadcast client.

```
D1(config)# interface g1/0/5
D1(config-if)# ntp broadcast client
D1(config-if)# exit
```

b. Configure the local time zone.

```
D1(config)# clock timezone EST -5
D1(config)# exit
Oct 29 20:17:00.150: %SYS-6-CLOCKUPDATE: System clock has been updated from 20:17:00
UTC Wed Oct 29 2025 to 15:17:00 EST Wed Oct 29 2025, configured from console by
console.
```

c. On D2, enter interface g1/0/5 and enable it to be an NTP broadcast client.

```
D2(config)# interface g1/0/5
D2(config-if)# ntp broadcast client
D2(config-if)# exit
```

a. Configure the local time zone.

```
D2(config)# clock timezone EST -5
Oct 29 15:20:08.139: %SYS-6-CLOCKUPDATE: System clock has been updated from 09:20:08
CST Wed Oct 29 2025 to 10:20:08 EST Wed Oct 29 2025, configured from console by
console.
D2(config)# exit
```

d. On D2, verify if there are any NTP clients or peers on the network by using the **show ntp associations** command.

```
D2# show ntp associations
```

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```
      address          ref clock      st  when  poll reach  delay  offset    disp
* 10.10.0.1         172.16.0.1       12    58    64  376     3.0   22.06     1.9
 * master (synced), # master (unsynced), + selected, - candidate, ~ configured
```

Notice that it has associated with R1 and recognizes it as the master.

e. Verify the status of NTP using the **show ntp status** command.

```
D2# show ntp status
Clock is synchronized, stratum 13, reference is 10.10.0.1
nominal freq is 119.2092 Hz, actual freq is 119.2062 Hz, precision is 2**18
reference time is E1D0954A.0BF64157 (16:08:26.046 EST Mon Jan 20 2020)
clock offset is 19.2737 msec, root delay is 4.96 msec
root dispersion is 53.60 msec, peer dispersion is 3.52 msec
```

Note: It may take a few minutes before the time is synchronized with the NTP server.

Step 4: Verify NTP on all devices.

Issue the **show clock detail** on all devices to validate that they are indeed synchronized.

```
D1# show clock detail
10:21:49.780 EST Wed Oct 29 2025
Time source is NTP

D2# show clock detail
10:21:55.365 EST Wed Oct 29 2025
Time source is NTP

A1# show clock detail
10:22:10.153 EST Wed Oct 29 2025
Time source is NTP

R1# show clock detail
10:22:15.160 EST Wed Oct 29 2025
Time source is NTP

R2# show clock detail
10:22:20.249 EST Wed Oct 29 2025
Time source is NTP

R3# show clock detail
10:22:27.772 EST Wed Oct 29 2025
Time source is NTP
```

The **show clock detail** commands were entered sequentially in the devices which explains the few seconds in between each command output. The output reasonably confirms that the clocks on these devices are all synchronized.

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